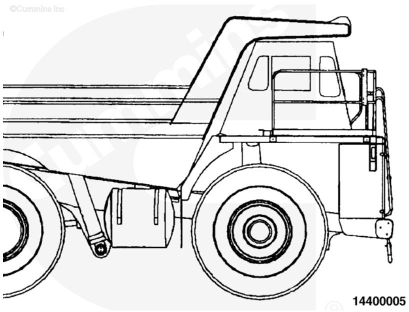
Trainings

General Information

⚠ CAUTION ⚠

Incorrect or insufficient break-in of the piston rings will lead to early oil consumption or high blowby complaints. Adherence to these run-in guidelines will allow the full durability of new pistons, liners, and rings.



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The engine test is a combination of the engine run-in period and the performance check. The engine run-in period procedure provides an operating period that allows the engine parts to achieve the final finish and fit. The performance check provides an opportunity to perform final adjustments needed to optimize the engine performance.

The engine run-in can be performed with an engine dynamometer. If a dynamometer is **not** available, the engine run-in **must** be performed in a manner that simulates a dynamometer test.

Check the dynamometer before beginning the test. The dynamometer **must** have the capability to test the performance of the engine when the engine is operating at the maximum rpm and horsepower range (full power).

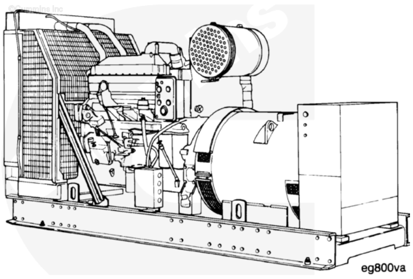
The crankcase pressure, often referred to as engine blowby, is an important factor that indicates when the piston rings and the valves have achieved the proper finish and fit to put the engine into a service application. Rapid changes of blowby, or values that exceed specifications more than 50 percent, indicate that something is wrong. The engine test **must** cease until the cause has been determined and corrected.

All engines **must** be run-in after a rebuild or a repair involving the replacement of one or more piston ring sets, cylinder liners, or pistons.

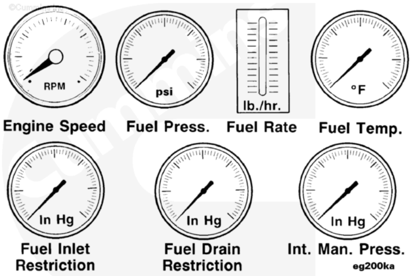
Before running the engine, make sure the cooling system is filled with the proper coolant and the lubricating oil system is filled and primed.

The in-service run-in is recommended when an engine dynamometer run-in can **not** be performed.

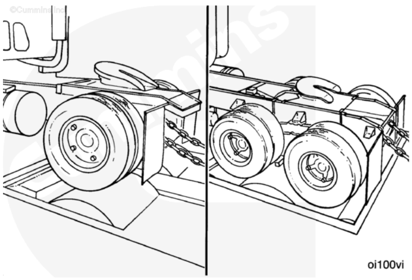
The majority of heavy-duty diesel applications will provide sufficient run-in under normal loaded operations. Light load, high rpm operation **must** be avoided during the run-in period.



Engine dynamometer run-in is the preferred method of engine run-in for engines that have been rebuilt out of chassis. It is **not** practical or recommended that an engine be removed from the application to conduct the run-in after a rebuild or cylinder repair has been performed in-chassis. There is no requirement or recommendation for an engine that has been run in and tested on an engine dynamometer to be run in again after it has been installed into the vehicle or equipment.



When it is **not** possible to load an engine immediately after rebuild or repair, the engine **must** be run in on a chassis dynamometer, portable dynamometer, or load bank.



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